

A Mathematical Theory of Value

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Abstract

We propose that *value* — the quantity that goal-directed agents create, destroy, and exchange — is a lawful structural quantity, in the same category as information, once stripped of its semantic clothing (morality, price, psychology). Following the method of Shannon (1948), we make one ruthless abstraction: value is the rate at which an agent converts a physical resource into goal-progress, relative to a frame fixed by the agent’s goal. From a single scale-invariance axiom we derive a logarithmic measure of value, $V = \sum_i k_i \ln e_i$. From the compounding dynamics of a resource that is reinvested we derive a *coding theorem of value*: the rate at which an agent can create value through a perception channel Y of the world X is bounded by the mutual information, $\Delta G \leq I(X; Y)$, achieved by Bayes-proportional allocation; and realized value decomposes exactly as available potential minus dissipation, $G = D(q \| r) - D(q \| p)$, identifying misalignment with measurable waste. For populations we show value is frame-relative while *price* is frame-independent, that the collective value throughput of a fleet is capped by the world’s entropy $\sum_a G_a \leq H(X)$, and that the fleet’s operating point is a Kelly portfolio over agents selected by an emergent price. A dynamical layer gives the equations of motion and an is/ought asymmetry — beliefs have a target the world supplies, goals do not — from which alignment emerges as a control-stability condition with a closed-form residual misalignment. We then test the single-frame laws on live language models: a capability ladder of local models on a frozen 100-item decision task obeys $\Delta G \propto I(X; Y)$ (Pearson $r = 0.996$), exhibits over-confidence as measurable dissipation that shrinks with capability, and confirms diversity-beats-redundancy; the fleet-pricing claim is reported with its honest boundary (no volatility-harvesting “demon” on a positively-correlated ladder, but pricing wins under a compute budget). None of the underlying mechanisms is individually new; the contribution is their unification under one substrate-grounded quantity, and the is/ought asymmetry that follows.

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1 Introduction

Information theory began with an abstraction that looked like a loss. Shannon (1948) threw away *meaning* and kept only the reduction of uncertainty; from that single move came a measure ($-\log p$), a unit (the bit), and hard limits (channel capacity, the coding theorems). The abstraction was productive precisely because it was ruthless: by refusing to model what a message *meant*, Shannon could state exactly what any communication system can and cannot do.

We attempt the analogous move for *value*. The everyday notion of value is a tangle of morality, market price, and human preference. We throw all three away and keep one structural residue: *value is the rate at which a goal-directed agent converts a scarce physical resource into progress toward its goal*. This is frame-relative — it is defined only relative to a goal, the way information is defined only relative to a prior and energy only relative to a reference frame — but relativity does not disqualify a quantity from being fundamental. It forces the theory to make the frame explicit. A theory of value is to agency what relativity is to motion: the quantity is frame-dependent; the laws relating frames are universal.

The motivation is not only foundational. If, as increasingly seems likely, near-future populations of artificial agents are each driven by an explicit objective, then a theory of value is not philosophy — it is the control theory for agent populations. To route, align, and govern many separated agents we need value to be a *measured* quantity with conservation laws and limits, not a verdict. This paper builds that quantity and tests it on real agents.

Contributions and honesty. We claim none of the underlying mechanisms as individually novel. The logarithm is Bernoulli’s and Kelly’s; the growth-rate-as-information-rate is Kelly’s; the price layer is Arrow–Debreu’s; the information geometry is Čencov’s and Amari’s. Our contribution is fourfold and we state it plainly (§11): (i) the *unification* of these accounts under one substrate-grounded quantity; (ii) the *capacity theorem* and *fleet ceiling* that the unification makes visible; (iii) the *is/ought asymmetry* as a structural property of agent dynamics; and (iv) an *empirical test* on live models. Throughout, each result is followed by the assumptions that make it bite, in Shannon’s spirit of naming the model.

Plan. Part I (statics) derives the measure, the capacity theorem, and the Second Law for a single agent. Part II (multi-agent) develops cross-frame price and the fleet capacity region. Part III (dynamics) gives the equations of motion and alignment-as-stability. Part IV (evidence) reports the synthetic and real-agent tests. Appendices collect the two proofs.

Part I

Statics: the measure, the limit, the Second Law

2 The measure: a logarithmic law of value

Setup. An agent holds a divisible resource of total size $E > 0$ (free energy, compute, budget, attention — any fungible substrate of action) and allocates it across K goal-relevant channels in amounts $e = (e_1, \dots, e_K)$, $\sum_i e_i = E$. Each channel i carries a *goal weight* $k_i \geq 0$ measuring how much progress toward the goal a unit of effective work on i yields. Let $V(e)$ be the value the allocation realizes. We ask what functional form V must take.

Axiom 1 (Diminishing returns). V is increasing and strictly concave in each e_i : the first unit of resource on a channel is worth more than the thousandth.

Axiom 2 (Additivity across independent channels). For channels whose contributions do not interact, $V(e) = \sum_i v(e_i; k_i)$.

Axiom 3 (Scale invariance). Rescaling the resource unit ($e_i \mapsto \lambda e_i$) changes total value only by an additive constant independent of the allocation: $v(\lambda e_i; k_i) = v(e_i; k_i) + c(\lambda, k_i)$.

Axiom 3 is the load-bearing one: it says value has no privileged zero of *scale*, exactly as information has no privileged base of logarithm. It is a Cauchy functional equation in disguise.

Theorem 1 (Logarithmic Value Law). *Under Axioms 1–3 with mild regularity, the value measure is, up to choice of unit,*

$$V(e) = \sum_{i=1}^K k_i \ln e_i \quad (1)$$

and the value-maximizing allocation of a fixed budget E is proportional to the goal weights, $e_i^* = E \hat{k}_i$ with $\hat{k}_i = k_i / \sum_j k_j$, giving

$$V^* = K \ln E - K H(\hat{k}), \quad H(\hat{k}) = -\sum_i \hat{k}_i \ln \hat{k}_i. \quad (2)$$

The proof (Appendix A) solves the Cauchy equation forced by Axiom 3. Two features deserve note. First, Shannon’s entropy H reappears *unbidden* as a penalty: a goal spread thinly over many channels (high $H(\hat{k})$) realizes less value than a focused one — *focus is worth $K H(\hat{k})$ nats of value*. Second, the logarithm here is forced by diminishing returns; §3 forces it a *second* time from compounding. A quantity whose form is over-determined by two independent routes is the signature of something fundamental, not a modeling convenience.

3 The capacity theorem: value-throughput is bounded by information-throughput

The repeated game. Now let value compound: value realized this round becomes resource next round (an agent that achieves its goal earns more budget, trust, or compute). Each round t the world settles into a state $s \sim q$; the agent splits its budget into fractions $b = (b_1, \dots, b_n)$, $b_i \geq 0$, $\sum_i b_i = 1$; and channel i returns a multiplier $o_i(s) \geq 0$ per unit committed. The budget updates multiplicatively, $E_{t+1} = E_t \sum_i b_i o_i(s_t)$, so $\log E_T = \log E_0 + \sum_t \log(\sum_i b_i o_i(s_t))$ and the long-run *value growth rate* is

$$G(b) = \mathbb{E}_{s \sim q} \left[\log \sum_i b_i o_i(s) \right]. \quad (3)$$

That the time-average growth of a multiplicative process is the *expected log* of its multiplier is the ergodicity argument of Peters (2019); it forces the logarithm again, now from dynamics rather than psychology. G is concave in b , so maximizing it is a well-posed convex program.

The horse race. Take the canonical structure in which states are the channels, committing to channel i pays only in state i , and returns are quoted against a *reference belief* r over states, $o_i = 1/r_i$. Then $G(b) = \sum_s q_s \log(b_s/r_s)$, maximized (Gibbs) by betting one’s model, $b_s = p_s$. With a correct model $p = q$,

$$G^* = \sum_s q_s \log \frac{q_s}{r_s} = D(q \| r). \quad (4)$$

Maximal value growth equals the divergence of the agent’s correct model from the baseline. An agent whose model is the baseline ($p = r$) grows value at rate zero: value creation is informational edge over the prevailing expectation, nothing more.

Side information. Equip the agent with a perception channel: a signal Y correlated with the world-state X , observed before allocating. Optimal play bets the posterior $b_s = p(s | y)$, with growth $G_Y = \sum_{x,y} p(x,y) \log \frac{p(x|y)}{r(x)}$. The gain over acting on the prior alone is exactly mutual information:

Theorem 2 (Coding Theorem of Value). *The incremental rate at which an agent can compound value by perceiving the world through a channel Y is at most the mutual information between world-state and perception,*

$$\boxed{\Delta G = G_Y - G_0 = I(X; Y)}, \quad (5)$$

and this bound is achieved by Bayes-proportional allocation. (Proof: Appendix B.)

This has the full structure of a Shannon coding theorem. *Achievability:* betting the posterior attains $\Delta G = I(X; Y)$. *Converse:* any $b \neq p(\cdot | y)$ loses exactly $D(p(\cdot | y) \| b)$ per round (Gibbs), so $I(X; Y)$ is an upper bound. An agent cannot create value faster than it can perceive the world; *value-throughput is bounded by information-throughput*. Perception capacity is the hard ceiling on value-generation rate — the value analog of channel capacity.

4 The Second Law of Value

Drop the assumption that the agent’s model is correct. Let q be reality, p the agent’s model, r the baseline; the agent bets $b = p$ but the world is drawn from q . Then

$$G_{\text{actual}} = \sum_s q_s \log \frac{p_s}{r_s} = \underbrace{D(q \parallel r)}_{\text{available potential}} - \underbrace{D(q \parallel p)}_{\text{dissipation (model error)}}. \quad (6)$$

Realized value is available potential minus dissipation from misalignment, every term in nats. Three consequences. (i) *Misalignment is measurable dissipation*: confident error ($D(q \parallel p) > D(q \parallel r)$) drives growth negative — a wrong, certain agent actively destroys value, while calibrated humility is value-preserving. (ii) *Learning is value-recovery*: $D(q \parallel p)$ is exactly the cross-entropy excess that predictive training minimizes, so the machine-learning objective *is* the minimization of value dissipation. (iii) *Alignment is a value law*: reducing $D(q \parallel p)$ is mechanically identical to reducing wasted value. We test (6) on live models in §10 (R2), where the weakest model’s over-confidence indeed drives realized growth negative.

Part II

Multi-agent: price and the fleet

5 Cross-frame value and the frame-independence of price

A single agent’s value lives in its own frame (q, p, r, k) . Two agents with different goals have values that are *not* cardinally comparable — the interpersonal-comparison impossibility of Arrow (1951), which we concede rather than circumvent. Yet they can still coordinate, exactly as economies coordinate non-comparable utilities: through a *price* on the shared resource.

Shadow price. For an agent with goal mass $K = \sum_i k_i$ and budget E , the marginal value of resource (the Lagrange multiplier on $\sum_i e_i = E$ in Theorem 1) is the *shadow price* $\lambda = K/E$. Two agents trade resource until their shadow prices equalize; the common λ is a scalar that lives in *neither* agent’s value frame. Hence:

Value is frame-relative; price is frame-independent. Value is a vector in an agent’s goal basis; price is the one scalar on which separated frames agree.

The invariant. The transformation between two agents’ value frames is a change of coordinates on the belief simplex. The unique (up to scale) Riemannian metric invariant under such sufficient-statistic reparametrizations is the Fisher–Rao metric (Čencov, 1982; Amari, 2016). It is the cross-frame invariant of the theory: distances in belief that all agents agree on, whatever their goals. Alignment between two goals is the cosine $\cos \theta_{ab} = \langle \hat{k}_a, \hat{k}_b \rangle$; its sign sets whether their interaction is positive- or negative-sum.

6 The fleet capacity region

Consider m agents acting on one world X and drawing on one resource pool.

The ceiling. Each agent’s growth is bounded by its own perception (Theorem 2), $G_a \leq I(X; Y_a)$. Jointly, the data-processing inequality gives the sum-rate bound

$$\boxed{\sum_{a \in S} G_a \leq I(X; Y_S) \leq H(X)} \quad \text{for any subset } S, \quad (7)$$

the collective value throughput of a fleet is capped by the entropy of the world it acts on. Headcount past perception-saturation buys nothing. *Perception diversity* (agents that perceive different slices of X) lifts $I(X; Y_S)$ toward $H(X)$; *redundancy* (agents perceiving the same slice) adds exactly zero. We confirm both on live models in §10 (R4).

The operating point. Under multiplicative dynamics the shared pool multiplies each round by $\sum_a w_a R_a$ for resource weights w_a , so the fleet is a *Kelly portfolio over agents*: for imperfectly-correlated agents, spreading resource and rebalancing raises the time-average growth (the volatility-harvesting effect of Kelly, 1956; Cover, 1991). The weights that achieve the optimum are selected by the resource *price* π : price moves resource to high-shadow-price agents until λ equalizes. Thus

the *alignment matrix* $M = [\cos \theta_{ab}]$ shapes the achievable region (a cooperation dividend where $\cos \theta > 0$, a conflict tax where $\cos \theta < 0$); the *price* selects the operating point on it.

This separates governance into two acts: *shaping* the region (alignment and perception design) and *choosing* the point on it (pricing). The pricing claim is the one we test most carefully (§10, R5), and where the demon’s precondition — genuine diversity — turns out to matter.

Part III

Dynamics: motion and alignment

7 The equations of motion and the is/ought asymmetry

The statics fix equilibria; the dynamics say how they are approached. Three objects evolve. *Beliefs* p flow toward the true law q by a natural-gradient (Fisher) descent on log-loss — the realized regret of this flow equals the cumulative dissipation $\sum_t D(q \| p_t)$ of (6), so *learning is value-recovery* dynamically as well as statically. *Prices* π flow toward market-clearing by tâtonnement (dual ascent on the resource constraint). *Goals* k flow under two forces: *control* (a principal pulling k_a toward a target k^*) and *selection* (compounding reweights resource toward higher-growth goals).

The organizing fact is an asymmetry. Beliefs and prices each flow toward a target *the world supplies*: reality provides q , the resource constraint provides market-clearing. Goals have *no* target the world supplies — reality contains no fact about what ought to be valued. So beliefs and prices are *learnable* (gradient flows toward a given target); goals are only *controllable* or *selectable*. This is Hume’s is/ought gap recovered as a structural property of the dynamics, and §8 shows it is the mathematical shape of the alignment problem.

8 Alignment as a stability condition

Let agent a have goal k_a and resource share w_a , with fleet effective goal $\bar{k} = \sum_a w_a k_a$. Control pulls each goal toward target: $\dot{k}_a|_{\text{ctrl}} = -\gamma(k_a - k^*)$. Selection follows the replicator $\dot{w}_a = w_a(G_a - \bar{G})$.

Summarize the environment’s reward landscape over goals to first order by its growth gradient $g := \nabla_k G$ — the direction in goal-space along which deviating from k^* *increases* resource capture.

Theorem 3 (Coupled-flow alignment). *By Price’s equation (Price, 1970), the selection drift of the mean is the trait–fitness covariance; with $G_a \approx \bar{G} + g^\top(k_a - \bar{k})$ it equals Vg , where $V = \text{Cov}_a(k_a)$ is the goal-dispersion matrix. Hence the effective goal obeys*

$$\dot{\bar{k}} = -\gamma(\bar{k} - k^*) + Vg, \quad (8)$$

with fixed point and residual misalignment

$$\boxed{\bar{k}^* = k^* + \gamma^{-1}Vg, \quad \|\bar{k}^* - k^*\| = \gamma^{-1}\|Vg\|,} \quad (9)$$

and the aligned fixed point is locally stable iff $\gamma > \lambda_{\max}(\partial(Vg)/\partial\bar{k})$.

The fleet does not in general settle at the target; it settles a distance $\|Vg\|/\gamma$ away — goal-dispersion times reward-pull over control gain. Perfect alignment holds iff $Vg = 0$: either the environment rewards exactly the target ($g = 0$), or there is no diversity to select among ($V = 0$), or control is infinite. The governance reading is sharp:

The cheap half of alignment is incentive design, not control. Driving $g \rightarrow 0$ (aligning what *pays* with what is *wanted*) removes the residual for *any* γ and *any* V , preserving the diversity that lifts the fleet ceiling (7); raising γ (brute-force oversight) merely opposes a drift it never removes. Spend first on g .

This is the actionable form of the is/ought asymmetry: because goals have no world-given target, they are governed by what pays (selection, via g) and what is imposed (control, via γ).

Part IV

Evidence

9 Synthetic validation

We first check that each closed-form prediction is reproduced by a Monte-Carlo world built to the theory’s assumptions. All five families pass to numerical tolerance (20/20 checks): E1, $\Delta G = I(X; Y)$ to $\sim 10^{-3}$ nats; E2, the Second-Law decomposition (6) exactly, including value going negative under confident error; E3, the fleet ceiling (7) with diversity lifting it and redundancy adding zero; E4, a Kelly-priced fleet beating ad-hoc allocation, including a “Shannon’s demon” built from anti-correlated agents that individually do not grow yet collectively do; E5, cumulative dissipation equal to regret, with a drifting world flooring dissipation at a positive value (a dynamical Second Law). These confirm the mathematics is self-consistent and correctly derived. They are, however, circular by construction: the worlds are drawn from the same distributions the formulas assume. The decisive test is on real agents.

10 Real agents

Instantiation. We take a frozen, pre-registered 100-item decision task in which each item has a ground-truth correct action drawn from $K = 7$ classes, and an agent’s output *is* an action. This

realizes the perception-then-act structure of Theorem 2 directly: the world-state X is the correct action, the perception Y_a is the action model a chooses. We set $r = q$ (the action marginal), so a no-signal agent grows value at rate 0. From each model’s run we form the confusion matrix $C_a[x, y] = \#(\text{gold} = x, \text{chosen} = y)$ and compute every quantity in nats with the same primitives used in §9: $I(X; Y_a)$ from the joint; the calibrated posterior $p_a(x | y) = C_a[:, y] / \sum_x C_a[:, y]$; and realized growth $\Delta G_a = \text{mean} \ln(p_a(x | y_a) / r(x))$. The agents are a weak→strong ladder of local models (1.5B, 3B, 7B parameters), greedy decoding. Posteriors are calibrated on a 48-item fit split and scored on a 52-item holdout; in-sample, the oracle posterior with $r = q$ makes $\Delta G = I$ an arithmetic identity (used only to confirm the units), so the empirical content is the ladder monotonicity, the out-of-sample tracking, and the fleet result.

R1: the bridge holds and rises with scale. Mutual information increases monotonically along the ladder and out-of-sample value-growth tracks it almost perfectly (Table 1; Pearson $r = 0.996$ between I and $\Delta G_{\text{holdout}}$). Value-throughput is information-throughput, on real behavior.

model	params	tool-acc	$I(X; Y)$	ΔG_{in}	ΔG_{hold}
A	1.5B	0.79	1.277	1.277	0.915
B	3B	0.87	1.561	1.561	1.242
C	7B	0.96	1.779	1.779	1.423

Table 1: R1 (nats). $I(X; Y)$ rises with capability; ΔG_{hold} tracks it ($r = 0.996$). The world entropy is $H(X) = 1.92$ nats, so model C captures 93% of it, model A 66%.

R2: over-confidence is dissipation, shrinking with capability. Reading the same point-predictions with a calibrated versus an over-confident posterior realizes different value; the gap is dissipation (Table 2). It is large for the weak model and shrinks with scale — a quantitative statement that weaker agents must be more humble. For model A, confident error drives realized growth *negative*, exactly the Second Law’s prediction (6).

model	G_{cal}	G_{over}	dissipated
A (1.5B)	+0.915	−3.255	4.17
B (3B)	+1.242	−0.863	2.11
C (7B)	+1.423	+0.731	0.69

Table 2: R2 (nats). Dissipation from false certainty falls as capability rises; the weakest agent’s over-confidence destroys value outright.

R3: value per joule. Mutual information *per second of compute* falls with scale ($0.74 \rightarrow 0.38 \rightarrow 0.30$ nats/s), as does I per billion parameters ($0.85 \rightarrow 0.52 \rightarrow 0.25$). The weakest model is the most efficient perceiver per unit compute — the value-theoretic reason a cheap reflex is worth running on a narrow task. (This is the $I/\text{compute}$ curve across model scale, not a within-model prompt ablation; see §13.)

R4: diversity beats redundancy. Two different models jointly perceive more of $H(X)$ than either alone; an identical greedy re-run adds exactly zero. The best diverse pair (B+C) reaches

$I = 1.869$ nats, $+0.090$ over C alone and closing most of the gap to $H(X) = 1.921$. This is the fleet-ceiling prediction (7) on real agents.

R5: pricing beats pooling, but only where there is diversity to price (reported honestly). On raw out-of-sample growth, the Kelly/price fleet ($+1.343$) beats the equal-weight ensemble ($+1.324$) but *does not beat the single best model* ($+1.423$). There is no Shannon’s demon here, and the theory says why: a capability ladder on the *same* task yields positively-correlated agents (R4 quantifies the residual diversity at $+0.09$ nats), so Kelly rebalancing has no anti-correlated volatility to harvest and the best agent dominates. This is the honest negative the synthetic E4 demon does not reproduce, because E4 was constructed with anti-correlated agents.

The negative is confined to the cost-blind axis. Value *per unit compute* inverts the ranking: model A yields 0.531 nats/s against C’s 0.243 , and a budget-aware price $\propto I_a/\text{cost}_a$ (the shadow price $\lambda = K/E$ of §5) achieves 0.393 nats/s, beating the best single model’s density. Pricing therefore pays exactly where §6 says it should: as the lever that chooses the operating point under a resource constraint, not as a free lunch that beats the best agent when compute is unlimited. The demon needs *perception diversity* — different slices of $H(X)$ — which a same-task scale ladder lacks by construction. We mark this as a falsifiable boundary, not a defeat.

11 Related work

Every component of this theory exists in some field; the contribution is the unification, the substrate grounding, and the is/ought asymmetry. We state this explicitly and concede each component to its source.

Expected utility and diminishing value. The axiomatic treatment of preference is von Neumann and Morgenstern (1944) and Savage (1954); our logarithmic measure (§2) coincides with Bernoulli (1738) and with unit-coefficient constant-relative-risk-aversion utility, and the diminishing-returns law it encodes is Weber–Fechner. We do not claim the form as novel; our departure is to *derive* it from scale invariance on a physical resource, to ground it in a substrate (free energy) supplying a unit and a cross-frame exchange rate that expected utility lacks, and to show it is forced a second time by compounding (Peters, 2019).

Information rate and log-optimal growth. The closest prior art, which we credit most carefully, is Kelly (1956): the exponential growth rate of wealth is an information rate, optimized by betting one’s beliefs; Breiman (1961) proved asymptotic optimality and Cover (1991) developed universal portfolios. Our capacity theorem (§3) is generalized Kelly, and the fleet operating point (§6) is a Kelly–Cover portfolio over agents. What we add is the reinterpretation of wealth as free energy (a physical rather than monetary substrate), the cross-frame price layer, the dynamics/alignment layer, and the claim that Kelly’s theorem is the single-agent monetary *special case* of a general law of value.

Reinforcement learning. “Value function” is precisely defined in RL (Bellman, 1957; Sutton and Barto, 2018) as expected cumulative reward; the relationship is complementary, not competitive. RL takes reward as given and maximizes it; our goal weights are the analog of reward, but we problematize where reward comes from and how it drifts (§7–8). Our theory sits beneath RL — a candidate account of what reward *is*.

Thermodynamics of computation and the free-energy principle. That information has thermodynamic cost is Landauer (1961) and Bennett (1982); Jaynes (1957) recast statistical mechanics as inference; England (2013) studied dissipation-driven adaptation; Friston (2010) casts agents as free-energy minimizers. Our belief dynamics (§7) is free-energy-principle-like and shares the infor-

mation geometry (Čencov, 1982; Amari, 2016). The distinction is precise: the free-energy principle is a theory of the perception/belief half — the “is” — whereas our value layer is the goal/value half — the “ought” — plus a multi-agent price economics the FEP does not contain. The seam between them is the is/ought asymmetry.

General equilibrium and social choice. Our price layer (§5) is Arrow and Debreu (1954); Debreu (1959) and its companion impossibility (Arrow, 1951). We use these rather than extend them: we concede that cardinal cross-agent comparison is impossible and route around it via an emergent price. The contribution is the synthesis with the information-theoretic value layer under one substrate, and the application to artificial-agent populations.

AI alignment. Instrumental convergence (Omohundro, 2008; Bostrom, 2014) we derive from the compounding dynamics of value (§7): goal-directions correlated with resource capture are selected regardless of terminal content. We recast alignment as a dynamical stability condition (§8), which to our knowledge is a new framing.

12 Discussion: governing a population of agents

The results compose into a control theory for populations of separated agents. *Per agent*, value-generation rate is capped by the perception mutual information $I(X; Y_a)$ (Theorem 2); supplying an agent the goal-relevant bits cheaply — raising effective I per unit compute — lifts its value ceiling without raising its deliberation cost, which is the formal reason a cheap reflex can match expensive deliberation on a narrow task (R3 measures this curve directly). *Across agents*, values are not cardinally comparable, so coordination must run through an emergent price on shared resource (§5), never a god’s-eye utility sum. *Fleet design* is then two levers: cultivate perception diversity to lift the entropy ceiling (7) (R4), and price resource to the best operating point (R5) — with the caveat, learned empirically, that pricing earns its keep only where genuine diversity exists. *Alignment*, finally, is incentive design before oversight (Theorem 3): spend first on making what pays equal what is wanted.

13 Limits

We name the assumptions that bound each claim. The capacity theorem assumes compounding (reinvested value) and is cleanest in the horse-race structure; general returns give a concave program without the closed-form $D(q \| r)$. The fleet ceiling’s exact achievable region is an open problem of network-information-theory difficulty. The cross-frame layer concedes that cardinal interpersonal comparison is non-canonical; we do not solve it. On the empirical side: the in-sample $\Delta G = I$ is an arithmetic identity, not evidence — only the ladder monotonicity, the out-of-sample tracking, and R5 are empirical; R3 is an $I/\text{compute}$ curve across model scale, not a within-model prompt ablation; the calibration in R2 is constructed from the confusion matrix, not a probability the model reported, so the dissipation measured is that of a stated belief over those predictions; and the whole test is one task on one corpus — evidence that the bridge holds there, not a universal law. Degenerate runs in which a model emits no action (a serving/plumbing failure, not a decision) are detected and excluded rather than reported as low- I capability. The alignment theorem is mean-field with a linearized landscape and a quasi-static dispersion V ; a fully coupled treatment, and an endogenous reward gradient g in a closed multi-agent world, remain open.

14 Conclusion

Stripped of morality, price, and psychology, value is a structural quantity with a measure ($\sum_i k_i \ln e_i$), a capacity limit ($\Delta G = I(X; Y)$), a Second Law ($G = D(q \| r) - D(q \| p)$), a frame-independent price, a fleet entropy ceiling, and an alignment-stability law. Shannon's own quantities — entropy, divergence, mutual information, the Fisher metric — reappear unbidden throughout, which is the strongest internal evidence that the abstraction is real rather than decorative. On live language models the single-frame laws hold to high precision, and the one fleet-level claim with a genuine precondition is reported with its boundary intact. The quantity is frame-relative; the laws relating frames are universal; and they are now testable on the artificial agents the theory was built to govern.

A Proof of the Logarithmic Value Law (Theorem 1)

By Axiom 2 it suffices to find the per-channel $v(e; k)$. Axiom 3 states $v(\lambda e; k) = v(e; k) + c(\lambda, k)$ for all $\lambda, e > 0$. Fix k and write $f(e) := v(e; k)$. Then $f(\lambda e) - f(e) = c(\lambda)$ is independent of e . Differentiating in λ at $\lambda = 1$, $ef'(e) = c'(1) =: k$, a separable ODE with solution $f(e) = k \ln e + \text{const.}$ Concavity (Axiom 1) fixes the sign $k \geq 0$ and rules out the additive constant's dependence on e . Summing over channels gives $V(e) = \sum_i k_i \ln e_i$. Maximizing under $\sum_i e_i = E$ with multiplier λ : $\partial_{e_i}(V - \lambda \sum_j e_j) = k_i/e_i - \lambda = 0 \Rightarrow e_i = k_i/\lambda$, and $\sum_i e_i = E$ gives $\lambda = K/E$ (the shadow price) and $e_i^* = Ek_i$. Substituting, $V^* = \sum_i k_i \ln(E\hat{k}_i) = K \ln E + \sum_i k_i \ln \hat{k}_i = K \ln E - KH(\hat{k})$ after normalizing $k_i = K\hat{k}_i$. $\square$

B Proof of the Coding Theorem of Value (Theorem 2)

Achievability. With side information $Y = y$ the optimal bet is the posterior $b_s = p(s | y)$ (Gibbs applied conditionally), giving conditional growth $\sum_s p(s | y) \log \frac{p(s|y)}{r(s)}$. Averaging over y ,

$$G_Y = \sum_y p(y) \sum_s p(s | y) \log \frac{p(s | y)}{r(s)} = \sum_{x,y} p(x, y) \log \frac{p(x | y)}{r(x)}.$$

Without side information the optimal bet is the prior $b_s = p(s)$, giving $G_0 = \sum_x p(x) \log \frac{p(x)}{r(x)} = D(p_X \| r)$. Subtracting,

$$\Delta G = G_Y - G_0 = \sum_{x,y} p(x, y) \log \frac{p(x | y)}{p(x)} = I(X; Y).$$

Converse. For any allocation $b(\cdot | y) \neq p(\cdot | y)$, the conditional growth loss is $\sum_s p(s | y) \log \frac{p(s|y)}{b(s|y)} = D(p(\cdot | y) \| b(\cdot | y)) \geq 0$ by Gibbs, with equality iff $b = p(\cdot | y)$. Hence no strategy exceeds $G_0 + I(X; Y)$, so $I(X; Y)$ is both achieved and an upper bound on ΔG . $\square$

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